

# Synoptic CB: Macrophyte Fluxes

2025 Samples

2026-03-16

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#Read in and collate all the files in the Processed Data folder

##Pull in Hydrago probe data

## Some flux IDs are missing from hydrago data

```
## [1] "GCW_TR_1_202509" "GCW_TR_2_202509" "GCW_TR_3_202509" "GCW_TR_4_202509"
## [5] "GCW_TR_5_202509" "GCW_WC_1_202509" "GCW_WC_2_202509" "GCW_WC_3_202509"
## [9] "GCW_WC_4_202509" "GCW_WC_5_202509" "GWI_WC_1_202503" "GWI_WC_1_202507"
## [13] "GWI_WC_2_202503" "GWI_WC_2_202507" "GWI_WC_3_202503" "GWI_WC_3_202507"
## [17] "GWI_WC_4_202503" "GWI_WC_4_202507" "GWI_WC_5_202503" "GWI_WC_5_202507"
## [21] "MSM_TR_1_202507" "MSM_TR_2_202507" "MSM_TR_3_202507" "MSM_TR_4_202507"
## [25] "MSM_TR_5_202507" "MSM_WC_1_202507" "MSM_WC_2_202507" "MSM_WC_3_202507"
## [29] "MSM_WC_4_202507" "MSM_WC_5_202507" "SWH_WC_1_202505" "SWH_WC_1_202509"
## [33] "SWH_WC_2_202505" "SWH_WC_2_202509" "SWH_WC_3_202505" "SWH_WC_3_202509"
## [37] "SWH_WC_4_202505" "SWH_WC_4_202509" "SWH_WC_5_202505" "SWH_WC_5_202509"
```

##Remove bad fluxes and calculate how many there were

## Number flux measurements removed: 15

## Number negative CO2 fluxes removed: 1

## Number unusually high starting value fluxes removed: 3

## Number messy fluxes removed: 11

## Percentage flux measurements removed: 8.82%

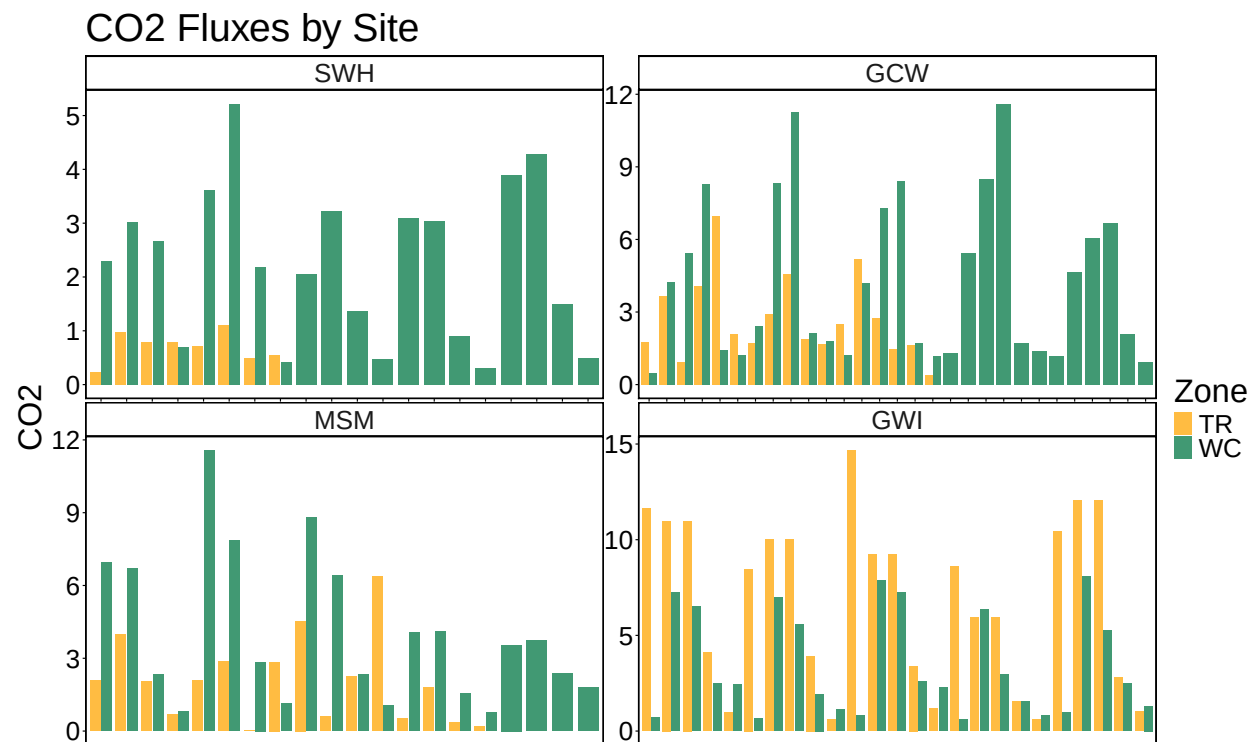
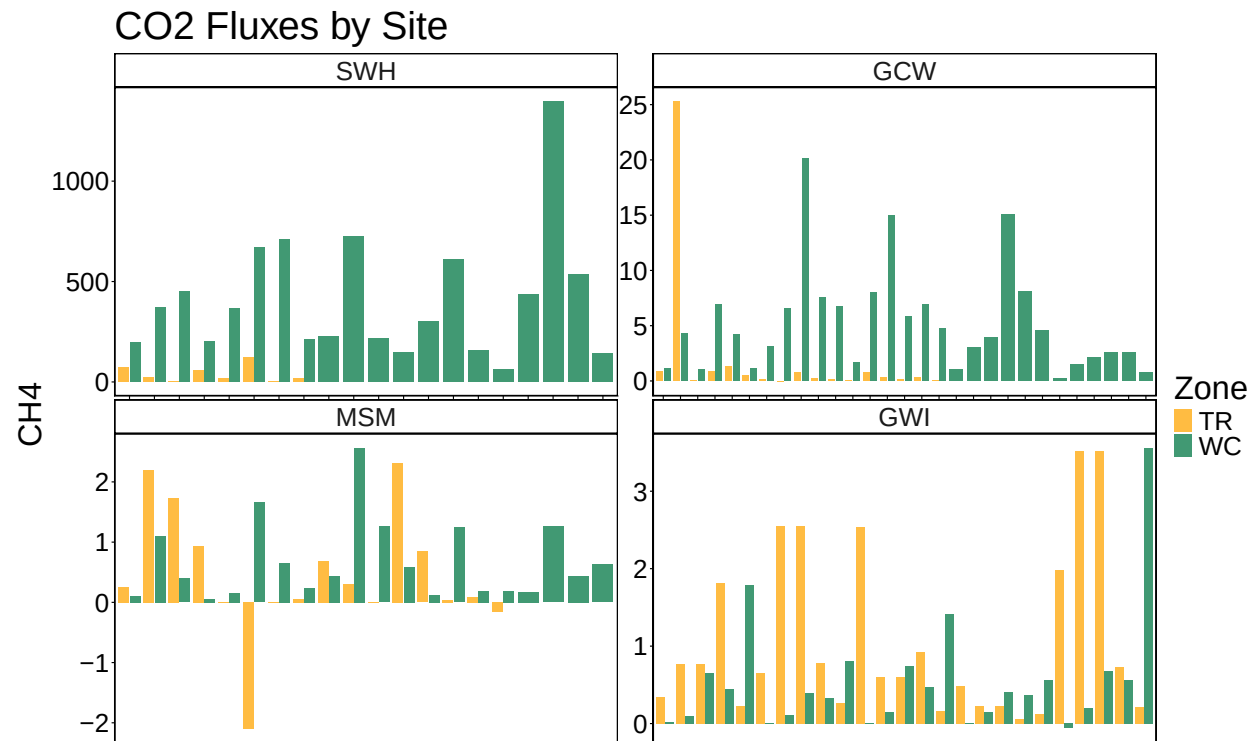
## Percentage negative CO2 fluxes removed: 0.59%

## Percentage unusually high starting value fluxes removed: 1.76%

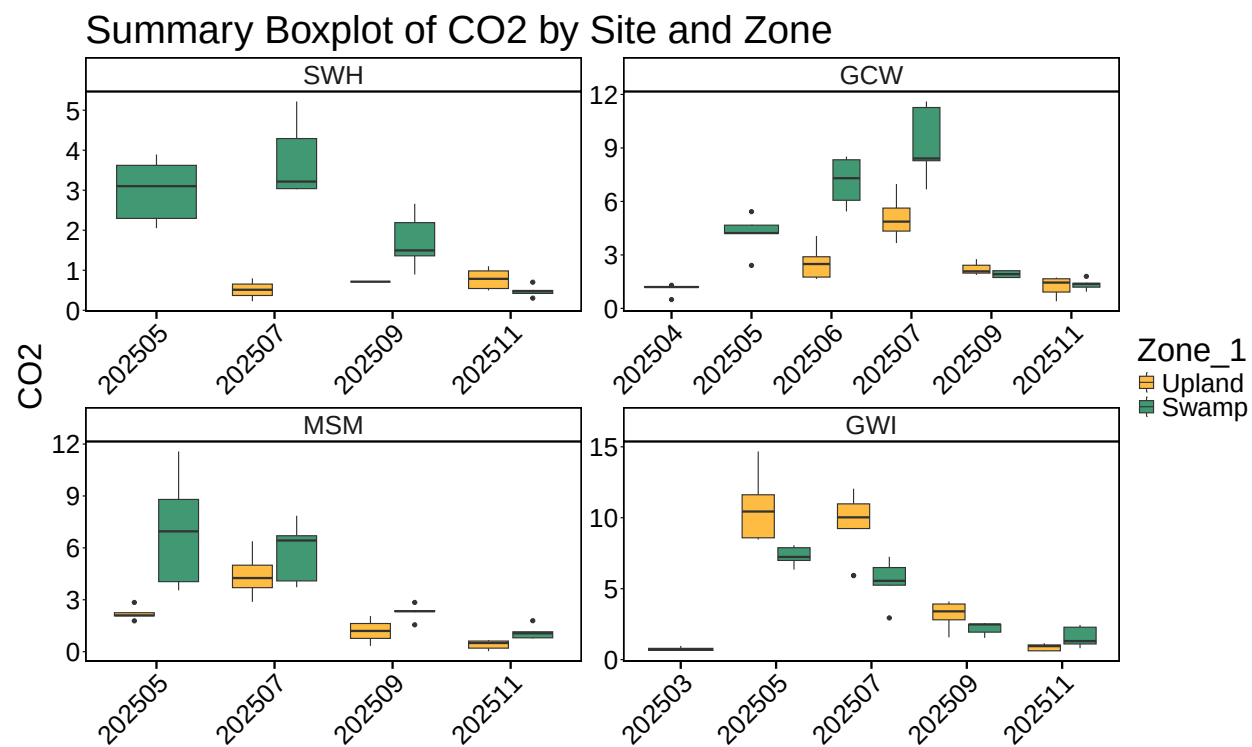
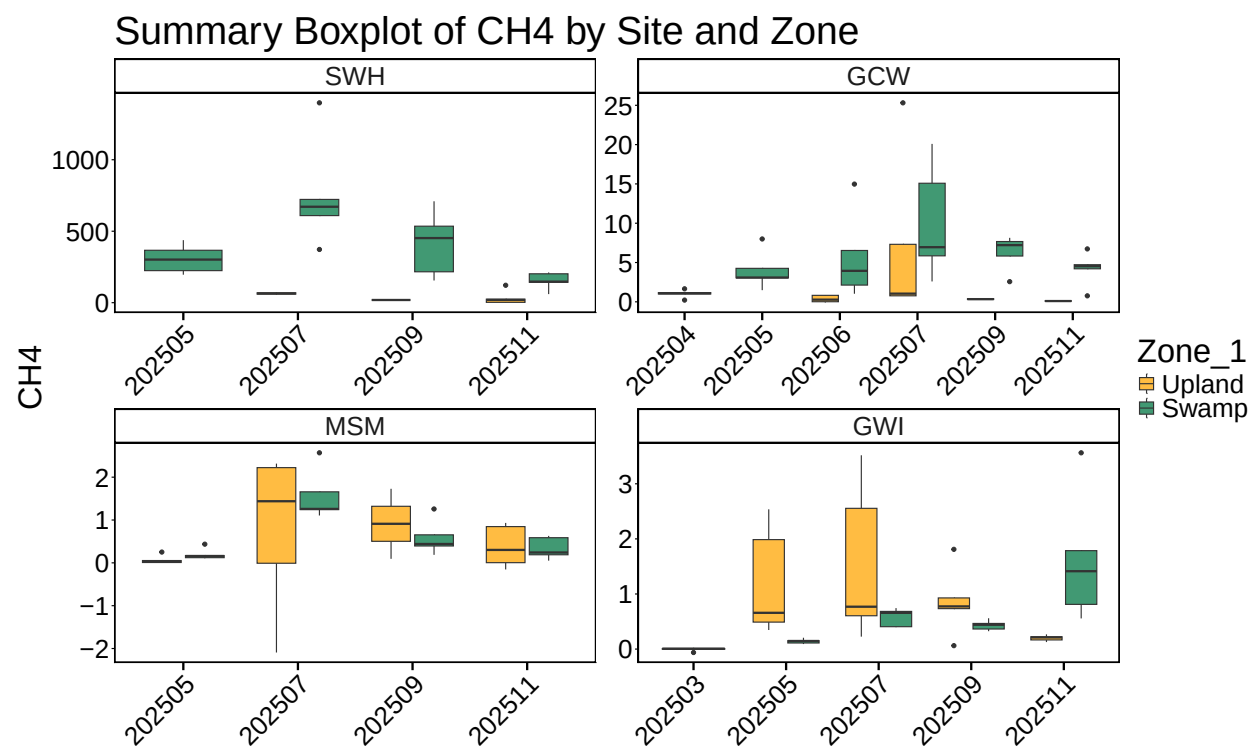
## Percentage messy fluxes removed: 6.47%

## 0.1 Write out files

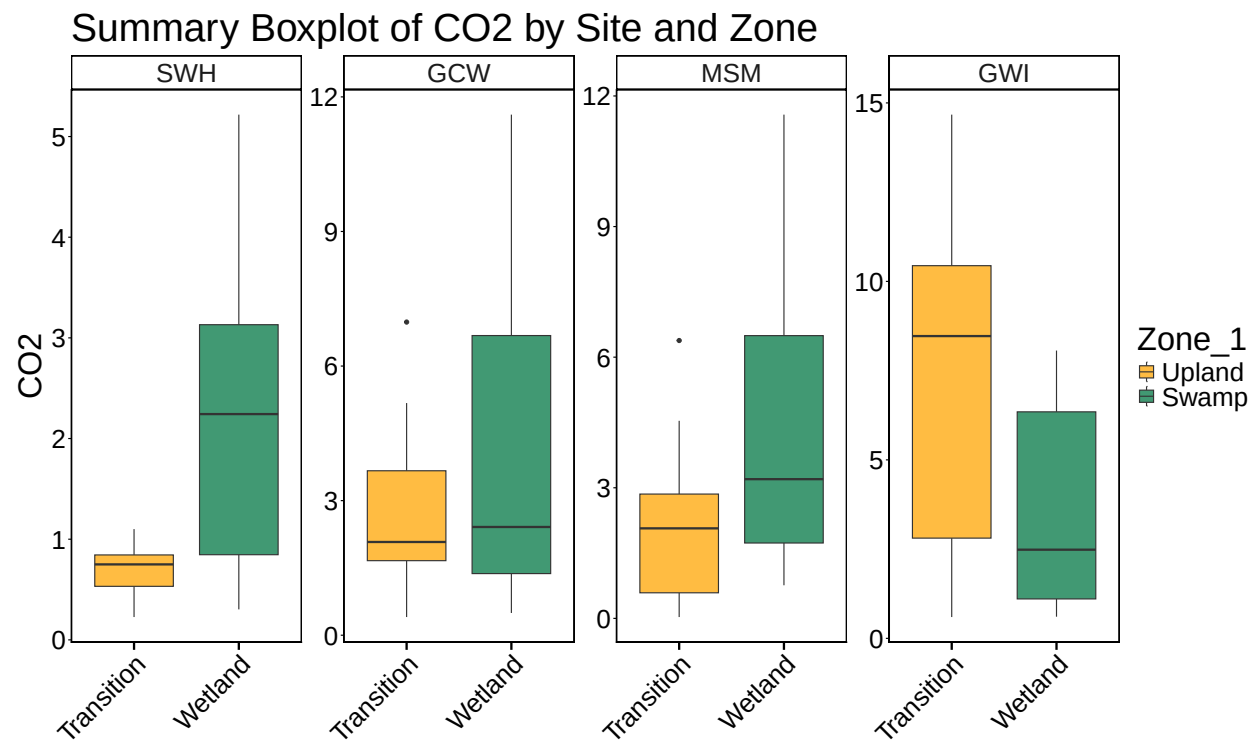
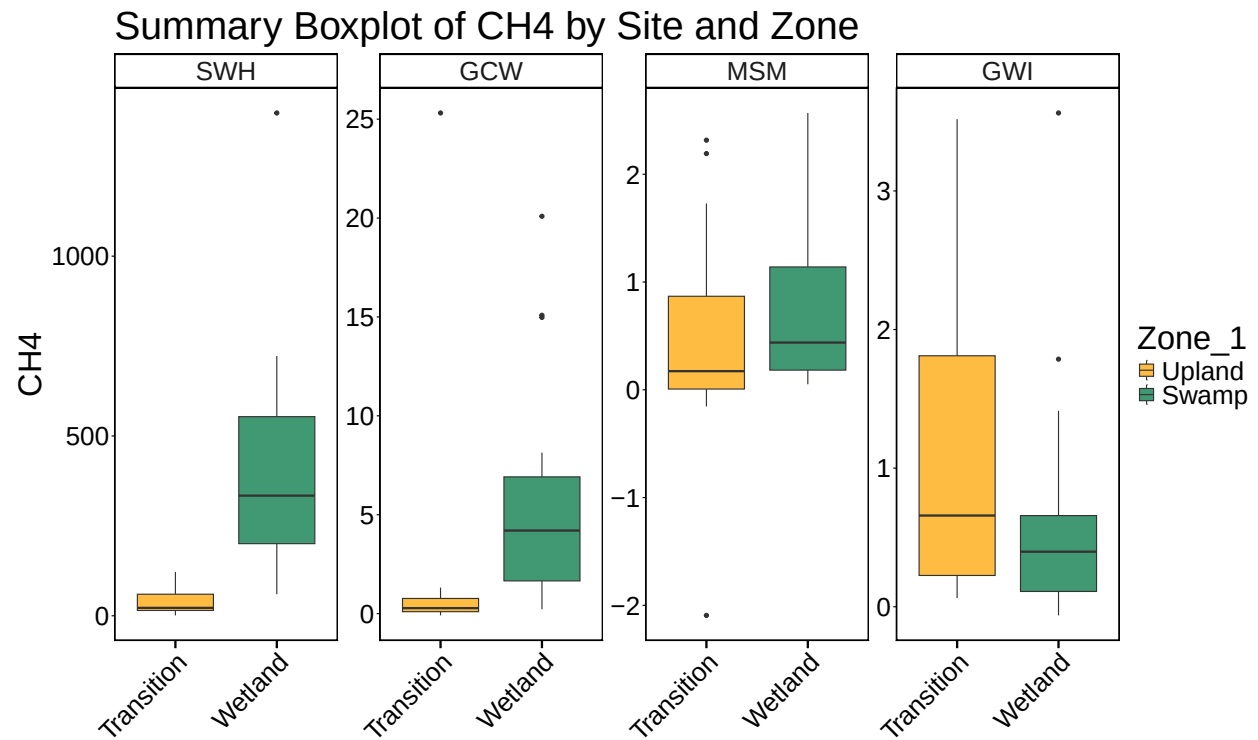
## 0.2 All Data by Year, Site and Zone



### 0.3 Boxplot summary by Date



#### 0.4 Boxplot summary by Site



##Calculate summary metrics

## 0.5 Summary Tables

Table 1: Summary Statistics

| Site       | Zone_1     | min_ch4    | max_ch4     | mean_ch4    | min_co2   | max_co2   | mean_co2  |
|------------|------------|------------|-------------|-------------|-----------|-----------|-----------|
| <b>SWH</b> | Transition | 1.0872692  | 121.977500  | 39.7087775  | 0.2266990 | 1.100404  | 0.7058451 |
| <b>SWH</b> | Wetland    | 59.9871778 | 1398.120534 | 406.5684788 | 0.3028537 | 5.217695  | 2.2376064 |
| <b>GCW</b> | Transition | -0.0901019 | 25.312906   | 1.8577613   | 0.4050037 | 6.980726  | 2.7131738 |
| <b>GCW</b> | Wetland    | 0.2215926  | 20.090639   | 5.1816997   | 0.4957703 | 11.604588 | 4.2294688 |
| <b>MSM</b> | Transition | -2.0931613 | 2.316486    | 0.4504229   | 0.0305177 | 6.382414  | 2.0786885 |
| <b>MSM</b> | Wetland    | 0.0514141  | 2.568636    | 0.6728730   | 0.7602931 | 11.571173 | 4.0361589 |
| <b>GWI</b> | Transition | 0.0621000  | 3.517843    | 1.0666977   | 0.5968420 | 14.671811 | 6.8158948 |
| <b>GWI</b> | Wetland    | -0.0623885 | 3.562404    | 0.5542746   | 0.6065226 | 8.062884  | 3.4680774 |